Venkata Lakshmi Parimala Pasupuleti

Full-Stack Platform Engineer · Python · FastAPI · React · TypeScript · Healthcare · FHIR · HL7 · HIPAA · Complex Workflow Platforms

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PROFESSIONAL SUMMARY

Senior full-stack platform engineer with 8 years shipping production applications across Python/FastAPI backends, React/TypeScript frontends, and complex multi-tenant data platforms. Built and owned healthcare data infrastructure end-to-end: **AI on FHIR** - a production platform translating natural-language queries into **FHIR/HL7 API calls** with **HIPAA-compliant RBAC**, reducing manual clinical retrieval by ~60%. At Entrata (8yr): designed APIs, data models, and integrations for 100K+ monthly active users across a complex multi-tenant healthcare-adjacent platform - income eligibility, rent calculations, PHI compliance, and document management workflows where data correctness is a regulatory requirement. Built **Leasel** - a FastAPI + LangGraph platform for complex configurable compliance workflows with structured extraction, hybrid RAG, and HITL routing. Strong on backend design (API design, data modeling, service integrations), frontend delivery (React/TypeScript), observability, and raising engineering quality through code review and mentorship. Open to relocation to San Francisco.

CORE COMPETENCIES

Python / FastAPI | JavaScript / TypeScript | React | Full-Stack Development | API Design & Data Modeling | System Integrations | FHIR / HL7 | HIPAA / PHI Compliance | Healthcare Data & Workflows | Complex Configurable Workflows | Multi-Tenant Architecture | Observability & Reliability | Production Services & APIs | AWS (S3 / Lambda / RDS) | Docker / Kubernetes | PostgreSQL / SQLite / Redis | CI/CD (GitHub Actions) | Data-Intensive Platforms | RabbitMQ / Async Processing | Technical Mentoring | Code Review & Engineering Practices

EXPERIENCE

Blue Horse Digital

Sep 2026 - Present

AI/ML Engineer (Volunteer) — Virginia, USA

- Building ML models and data analysis pipelines; integrating AI components with backend/frontend teams; surfacing business insights into application-facing APIs and reporting layers.

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern

- Built an agentic mission-planning system as a LangGraph stateful graph with FastAPI backend and MCP server; implemented RAG layer over regulatory corpus with deterministic guardrails; delivered end-to-end from data modeling through deployment.

George Mason University

Aug 2024 - Aug 2025

Research Associate / Graduate Teaching Assistant — Virginia, USA

- Built multi-source prediction models (Python, R) on 190+ country datasets - R^2=0.977; applied anomaly detection across 3 research pipelines; fine-tuned a classification model (HuggingFace + PyTorch), deployed inference API on Quart.
- Delivered technical workshops to 600+ students covering Python programming, database design, and SQL across MIS 303 sections.

Entrata - PropTech / Real Estate SaaS

Sep 2018 - Aug 2024

Senior Software Engineer — Lehi, UT (Remote)

- **Shipped 20+ full-stack production modules** across 120+ Agile sprints - Python/PHP backends, PostgreSQL data models, JavaScript/jQuery frontends - powering complex configurable eligibility, financial, and document workflows for 100K+ monthly active users.
- **Owned regulated data pipeline and API design for 7 years** - HUD income-limit ingestion (few thousand rows → ~180,000 domain records); designed validation, reconciliation, and rollback logic; zero compliance violations across the ownership period.
- **Built RabbitMQ-based async processing platform** for gross-rent workflow automation - retry logic, idempotency guarantees, and dead-letter queue handling; owned through production incidents and annual schedule changes.
- **Led AWS S3 document integration with 100% PHI compliance** - Libsodium-encrypted pipelines, multi-tenant data isolation at the schema layer, HIPAA-compliant architecture; optimized 200+ SQL queries (30% performance gain).
- Led microservices redesign using Docker (50% scalability improvement, 30% deployment time reduction); mentored 5+ junior engineers; conducted 300+ peer code reviews; built internal debugging tools cutting debug time by 40%.

Entrata India - Db Xento Systems

Jan 2017 - Sep 2018

Associate / Trainee Software Engineer — Pune, India

- Built property management and compliance modules (PHP, Python, Java, PostgreSQL); led AWS S3 cloud migration; income limits import (1.6M HUD/Tax Credit records across 5 years); created 56 state-specific housing certification documents.

PROJECTS

AI on FHIR: Healthcare Data Integration Platform

FHIR · HL7 · HIPAA · React · Clinical NLP

Built a **full-stack healthcare data platform** translating natural-language queries (EN/ES) into **FHIR/HL7 API calls** using LLMs, BERT, and embeddings - reduced manual clinical retrieval time ~60%. **HIPAA-compliant RBAC** at the data layer: PHI access enforced per role, not per request. React frontend with accessible query interface; Flask backend with multilingual clinical NLP (spaCy + BERT); containerized with Docker. Integrates with structured medical records across clinical workflows.

Flask · React · FHIR APIs · HL7 · HIPAA RBAC · LLMs · BERT · spaCy · Embeddings · Docker · 2024

Leasel: Complex Configurable Workflow Platform

FastAPI · LangGraph · Complex Workflows · Data-Intensive

FastAPI + LangGraph platform for complex configurable compliance workflows - structured extraction from unstructured documents, hybrid RAG retrieval (TF-IDF + semantic reranking), 8-agent orchestration with conditional routing and HITL escalation. **Data-intensive**: multi-source knowledge repository integrating regulatory documents, system settings, historical incidents, and live data. Observability built in: every agent transition traced, every decision cited to source. Achieved 93.6% precision on 1,240 test cases with CI regression gate.

Python · FastAPI · LangGraph · SQLite · TF-IDF · Semantic Reranking · OCR · CI/CD · 2026

AgentProbe: Open-Source Platform Engineering Toolkit

FastAPI · Observability · CI/CD · Multi-Provider

15-endpoint **FastAPI platform** with eval engine, model routing across OpenAI/Anthropic/Groq/Ollama, CI-integrated regression harness, and online + offline evaluation loops. Built for reliability and developer workflow improvement - golden datasets, automated regression gates, latency and performance monitoring. Zero vendor dependency; fully local. Live: github.com/venkatalakshmiparimala/AgentProbe

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) · 2025

EDUCATION

George Mason University **- M.S. Data Analytics Engineering** May 2026

GPA: 3.87 · Virginia, USA · Coursework: ML systems, NLP, cloud infrastructure, statistical modeling, deep learning

Jawaharlal Nehru Technological University **- B.Tech Computer Science & Engineering** 2017